rotation and, 7–8
vector analysis, 10, 11–12, 16
Foucault pendulum, 28, 76
Four-vector momentum, 105
Four-vectors, xiv–xv, 65–66, 102–109
Future (time-like interval), 99–100, 101

Galilean transformation, 52, 53, 54
Galileo, xii–xiii, 27–28
Gamma (τ)-meson, 40
General Theory of Relativity, xv, 50
Geographical differences, effect on physical laws of, 25
Geometry
of curved space, 117–125, 128–129
of space-time. *See* Space-time
Gravitation, laws of
change of scale, effect of, 27
Einstein's theory of, 141–144
equivalence principle and, 129–136
Newton's laws of, 111, 142
reflection symmetry in, 40
Gravitational field
motion law in, 140, 141–144
speed of clocks in, 131–136, 140–141

Huygens, Christian, 51

Inelastic collision, 87–88, 89
Inertia, 67, 88
Intervals, space-time, 97–99
Invariance. *See* Symmetry
Invariant. *See* Scalar

Kinetic energy, 20, 68–69, 88–91
K-mesons, 91

Least action, principle of, 141
Left-handedness, 31–39
Leighton, Robert B., xi
Light
distance measured by speed of, 98
photon of, 108–109
representation in space-time, 96, 98
velocity of, xiii, 52–53, 56–57, 75, 98, 100
wavelength of, variations in, 27
Lorentz, Hendrik Antoon, xii, xiii, xv, 58
Lorentz transformation
consequences of, 79–83
contraction, 63
laws of mechanics under, 66
as rotation in time and space, 65–66, 102, 105
simultaneity, 63–64
space-time relationships implied by, 93–95, 98
twin paradox, 77–79

Magnetic fields
cobalt experiment, 41–42, 45
electromagnetic field, Maxwell's equations of. *See* Electrodynamics, Maxwell's laws of